

Logic Handout

September 15, 2025

Exercise Group 1

In each exercise below, there are two statements. Form their conjunction AND disjunction, and determine if the resulting statement is true or false.

- 1) Some roses are red. Some violets are blue.
- 2) September has 30 days. There are eight days in a week.
- 3) 3 is less than 4. 4 is less than 3.
- 4) Napoleon was an emperor of Japan. Marco Polo discovered America.
- 5) $2 + 2 = 5$. $6 + 3 = 9$.
- 6) $7 + 4 = 11$. $8 + 0 = 9$.
- 7) $1 + 1 = 2$. $1 \times 1 = 2$.
- 8) $5 + 1 = 5$. $5 \times 1 = 6$.

Exercise Group 2

Form negations of the following statements.

- 1) The grass in my yard is green.
- 2) This rose is black.
- 3) I am tired.
- 4) The temperature of the sun is more than 1000 degrees.
- 5) Geometry is interesting.
- 6) $5 + 4 = 9$.
- 7) $6 + 3 = 7$.
- 8) $x = 2$ and $y = 3$.
- 9) $x = 2$ or $y = 3$.
- 10) All pigs are fat.
- 11) Some Americans are liars.

Exercise Group 3

Give the negation of each statement.

- 1) All men in Georgia are at least 6 feet tall.
- 2) All students in France like Geometry.

- 3) All students in Russia hate Geometry.
- 4) Every rectangle is a square.
- 5) Every triangle is equilateral.
- 6) All gadgets are widgets.
- 7) All miggles are maggles.
- 8) For every pair of numbers a and b , $a + b = b + a$
- 9) Every circle has a center.

Exercise Group 4

Give the negation of each statement.

- 1) Some men play golf.
- 2) Some boys play football.
- 3) There is at least one equilateral triangle.
- 4) For some number x , $x^2 + 2 = 5$.
- 5) There is at least one number x such that $x + 3 = 7$.
- 6) There is at least one number x such that $x^2 + 2 = 0$.
- 7) Some dogs have fewer than 4 legs.

- 8) Some numbers are negative.
- 9) Some freshmen always tell the truth.
- 10) Some people work hard.

Exercise Group 5

Give the negation of each statement.

- 1) Today is Monday and the weather is cold.
- 2) $x = 2$ and $y = 5$.
- 3) $2 \times 2 = 4$ and $3 \times 5 = 15$.
- 4) $\triangle ABC$ is isosceles and $\triangle ABC$ is a right triangle.
- 5) The point A is on the line l and the point B is on the line m .
- 6) All cats are black and all men are mortal.
- 7) All cats are black and some birds are blue.
- 8) Some rectangles are squares and some triangles are equilateral.
- 9) Some widgets are gadgets and no wiggles are waggles.

Exercise Group 6

Give the negation of each statement.

- 1) I will go to the game or I will go to the movie.

- 2) I will study hard or I will fail.
- 3) We will start at 6:30 or we will be late.
- 4) A donkey has three legs or a rabbit has long ears.
- 5) $2 + 2 = 5$ or $2 + 2 = 6$.
- 6) $3 \times 7 = 21$ or $4 \times 5 = 20$.
- 7) Billings is right or Jones is wrong.
- 8) Some will work or all will starve.
- 9) Every triangle is isosceles or every triangle is right.
- 10) No roses are red or some violets are blue.

Exercise Group 7

Put the following statements in an "if α then β " form so that α implies β .

- 1) If I eat strawberries I get hives.
- 2) The picnic will be canceled in case of rain.
- 3) We will win the game if Jones plays.
- 4) Numbers larger than 5 are positive.
- 5) $x^2 = y^2$ because $x = y$.

- 6) $3x^2 - 2 = 10$ when $x = 2$.
- 7) A geometric figure which is a square is not a circle.
- 8) Squares are rectangles.
- 9) The area of a rectangle with side lengths a and b is ab .
- 10) With a million dollars, I'd eat steak for breakfast.
- 11) People who don't study mathematics don't have good sense.
- 12) When it rains my rheumatism bothers me.
- 13) When I study I do well in school.
- 14) People who do not vote are poor citizens.
- 15) Two lines which intersect are not parallel.
- 16) I know that $x = 2$ because $3x = 6$.
- 17) Satisfaction guaranteed or your money back.
- 18) I know that $3x = 6$ because $x = 2$.
- 19) I will not go to the party if Jimmy is coming.
- 20) This shirt will not shrink or your money will be refunded.

- 21) This shirt will shrink or your money will be refunded.
- 22) To remain free we must vote in elections.
- 23) The crops will be saved if we have rain this week.
- 24) The crops will be saved only if we have rain this week.
- 25) I will go to the dance if Joe asks me.
- 26) I will go to the dance only if Joe asks me.
- 27) It will rain tomorrow if it turns cold.
- 28) It will rain tomorrow only if it turns cold.

Exercise Group 8

In Exercises 1 through 7, a statement $\alpha \rightarrow \beta$ is given which is to be accepted as true. What conclusion can you draw if (a) is true? if (b) is true?

- 1) If Fred is on the team he must go to bed early.
(a) Fred goes to bed early. (b) Fred does not go to bed early.
- 2) If a man is a postman he has foot trouble.
(a) This man has perfect feet. (b) That man has foot trouble.
- 3) You will be given a ticket if you are a sophomore.
(a) Jim was not given a ticket. (b) Joe was given a ticket.
- 4) You will be given a ticket only if you are a sophomore.
(a) Jean was given a ticket. (b) George is a sophomore.
- 5) If two triangles have equal bases and heights then they have equal areas.
(a) These triangles have equal areas. (b) These triangles do not have equal areas and they do have equal heights.

- 6) The house will burn down if we don't get some water.
(a) We did not get any water. (b) We did get some water.
- 7) If a man is beheaded in June then he will die before July 4.
(a) This man died before July 4. (b) Smith was not beheaded in June.

Is the reasoning in Exercises 8 through 18 correct?

- 8) All men are mortal. Socrates is a man.
Conclusion: Socrates is mortal.
- 9) All widgets are gadgets. This is a gadget.
Conclusion: This is a widget.
- 10) All widgets are gadgets. This is not a gadget.
Conclusion: This is not a widget.
- 11) Only widgets are gadgets. This is a gadget.
Conclusion: This is a widget.
- 12) Only widgets are gadgets. This is a widget.
Conclusion: This is a gadget.
- 13) Every equilateral triangle is isosceles. $\triangle ABC$ is isosceles.
Conclusion: $\triangle ABC$ is equilateral.
- 14) Parallel lines do not intersect. Lines r and s intersect.
Conclusion: Lines r and s are not parallel.
- 15) All of these dresses will shrink if washed in hot water. This dress did shrink.
Conclusion: It was washed in hot water.
- 16) All sweet food has honey in it. This food is sweet.
Conclusion: It has honey in it.
- 17) All sweet food has honey in it. This food doesn't have honey in it.
Conclusion: It is not sweet.
- 18) All sweet food has honey in it. This food is not sweet.
Conclusion: It does not have honey in it.